

Commerce XI Maths App

Let's Remember

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 12

Question 1 - MCQ · PG-34 · LR-1

Range equals

Options

- A. largest value – smallest value
- B. largest value + smallest value
- C. $Q_3 - Q_1$
- D. variance

Correct answer: largest value – smallest value

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives largest value – smallest value.

Question 2 - MCQ · PG-34 · LR-2

Interquartile range equals

Options

- A. $Q_3 - Q_1$
- B. $Q_3 + Q_1$
- C. $(Q_3 - Q_1)/(2)$
- D. Q_2

Correct answer: $Q_3 - Q_1$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $Q_3 - Q_1$.

Question 3 - MCQ · PG-34 · LR-3

Quartile deviation equals

Options

- A. $(Q_3 - Q_1)/(2)$
- B. $Q_3 - Q_1$
- C. $(Q_1)/(2)$
- D. $(Q_2)/(2)$

Correct answer: $(Q_3 - Q_1)/(2)$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $(Q_3 - Q_1)/(2)$.

Question 4 - MCQ · PG-34 · LR-4

Variance for raw data is the mean of

Options

- A. squared deviations from $\bar{x}$
- B. observations
- C. absolute class limits
- D. quartiles

Correct answer: squared deviations from $\bar{x}$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives squared deviations from $\bar{x}$.

Question 5 - MCQ · PG-34 · LR-5

Standard deviation equals

Options

- A. $\sqrt{\text{Var}(X)}$
- B. $\text{Var}(X)^2$
- C. $-\sqrt{\text{Var}(X)}$
- D. $(\text{Var}(X))/(2)$

Correct answer: $\sqrt{\text{Var}(X)}$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $\sqrt{\text{Var}(X)}$.

Question 6 - MCQ · PG-34 · LR-6

For frequency data, N equals

Options

- A. Σf_i
- B. Σx_i
- C. $\Sigma f_i x_i$
- D. $\bar{x}$

Correct answer: Σf_i

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives Σf_i .

Question 7 - MCQ · PG-34 · LR-7

For frequency data, $\bar{x}$ equals

Options

- A. $(\sum f_i x_i)/(N)$
- B. $(\sum x_i)/(N)$
- C. $(\sum f_i)/(N)$
- D. $(N)/(\sum f_i x_i)$

Correct answer: $(\sum f_i x_i)/(N)$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $(\sum f_i x_i)/(N)$.

Question 8 - MCQ · PG-34 · LR-8

Under step deviation, u equals

Options

- A. $(X - A)/(h)$
- B. $X - A$
- C. $(X)/(h) + A$
- D. $(h)/(X - A)$

Correct answer: $(X - A)/(h)$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $(X - A)/(h)$.

Question 9 - MCQ · PG-34 · LR-9

Under change of scale, $\text{Var}(X)$ equals

Options

- A. $h^2 \text{Var}(u)$
- B. $h \text{Var}(u)$
- C. $(\text{Var}(u))/(h^2)$
- D. $A \text{Var}(u)$

Correct answer: $h^2 \text{Var}(u)$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $h^2 \text{Var}(u)$.

Question 10 - MCQ · PG-34 · LR-10

The combined mean is a

Options

- A. size-weighted mean of group means
- B. simple average always
- C. difference of means
- D. sum of S.D.s

Correct answer: size-weighted mean of group means

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives size-weighted mean of group means.

Question 11 - MCQ · PG-34 · LR-11

Coefficient of variation equals

Options

- A. $100 \times (\sigma)/(\bar{x})$
- B. $100 \times (\bar{x})/(\sigma)$
- C. $(\sigma^2)/(\bar{x})$
- D. $(\bar{x})/(100)$

Correct answer: $100 \times (\sigma)/(\bar{x})$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $100 \times (\sigma)/(\bar{x})$.

Question 12 - MCQ · PG-34 · LR-12

A smaller coefficient of variation indicates

Options

- A. greater consistency
- B. greater variability
- C. larger range always
- D. negative variance

Correct answer: greater consistency

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives greater consistency.